MIL-S-20708/93B(MU)
26 July 1971
SUPERSEDING
MIL-S-20708/93A(MU)
12 January 1967

MILITARY SPECIFICATION

SYNCHROS, 400 Hz, SIZE 15

1. SCOPE

1.1 This specification covers the detail requirements for 115 volt, 400 Hz, size 15 synchros, types 15TX4a, 15TR4a, 15TR4b, 15CDX4a, 15CT4a, 15CX4, 15CX4a, and 15TDX4a. (See 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

TT-I-558	Ink, Marking Stencil, Opaque, for Nonporous Surfaces (Metals, Glass, etc.)

Military

MIL-V-173	Varnish, Moisture-and-Fungus Resistant (For the treatment of Communications, Electronic, and Associated Electrical Equipment)
MIL-S-20708	Synchros, General Specification For

STANDARDS

Military

MS 17186	Washer, Drive (Synchro)
MS 17187	Nut, Plain, Hexagon
MS 35276	Screw, Machine, Drilled, Fillister Head, Slotted, Corrosion Resisting Steel, Passivated, UNF-2A
MS 35338	Washer, Lock-Spring, Helical, Regular (Medium) Series

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 5990

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3. REQUIREMENTS

3.1 Certification.- The contractor shall certify that the synchros furnished under this specification have passed the qualification tests in accordance with MIL-S-20708 and the exceptions specified herein.

3.2 Design and construction.- Synchros furnished under this specification shall conform with the design, construction and physical dimensions specified in Figures 1, 2, and 3.

3.2.1 Housings.- Not applicable.

3.2.2 Slip rings.- Not applicable.

3.3 General requirements.- All requirements shall be in accordance with MIL-S-20708 except as otherwise specified herein. In case of conflict between the requirements of MIL-S-20708 and this specification, the requirements of this specification shall govern.

3.3.1 Impedance.- Not applicable.

3.3.2 Altitude.- Not applicable.

3.3.3 Vibration.- Synchros shall withstand mechanical vibrations of 0.10 $\pm$ 0.02 inch amplitude up to 60 cycles per second perpendicular and parallel to the shaft axis for a period of 8 hours $\pm$ 15 minutes without mechanical damage or loosening of parts, and with no greater degradation in performance than specified in MIL-S-20708 following conformance to the vibration requirement.

3.3.4 Shock.- Synchros shall be capable of withstanding shock blows of 100 G. After testing, there shall be no damage or loosening of parts and synchros shall meet performance requirements specified in MIL-S-20708 following shock testing.

3.3.5 Ambient temperature.- Synchros shall be capable of storage in ambients ranging from -62°C to 100°C. Synchros shall be capable of operating in ambient temperatures ranging from -55°C to 85°C. Synchros shall meet the applicable requirements in Table I at the specified ambient temperatures of MIL-S-20708, except the high ambient temperature shall be 85°C in lieu of 125°C.

3.3.6 Endurance.- All synchros shall be capable of continuous operation for 1000 $\pm$ 10 hours without failure or undue wear. After testing, all synchros shall meet the applicable requirements of MIL-S-20708. The high ambient temperature shall be 85°C in lieu of 125°C.

3.4 Performance requirements.- The performance requirements shall be as specified in Table I.

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TABLE I. Synchros, 400 Hz, Size 15

REQUIREMENT	UNITS	TYPES							
		CT4a	TDX4a	CDX4a	TR4a	TR4b	CX4	CX4a	TX4a
Primary voltage (nominal)	Volts	90	90	90	115	115	115	115	115
Primary current (± 10%)	Milliamps	20	190	90	225	115	75	75	225
Primary power (± 20%)	Watts	.50	3.3	1.4	2.7	---	1.1	1.1	2.7
Transformation ratio (± 1%)	---	.735	1.154	1.154	.783	.783	.783	.783	.783
Electrical error (max)	Minutes	8	8	8	8	6	8	8	8
Torque gradient (± .002)	Oz-in/deg	---	.014	---	.016	.03	---	---	.016
Receiver error (max)	Minutes	---	---	---	60	45	---	---	---
Null voltage total (max)	Millivolts	75	---	125	---	---	125	125	---
Fundamental component of null voltage (max)	Millivolts	40	---	75	---	---	75	75	---
Friction torque	Oz-in	.05	.05	.05	---	---	.05	.05	.05
(max @ -55° ± 5°C	Oz-in	0.1	0.1	0.1	---	---	0.1	0.1	0.1
Temperature rise (max)	°C	45	65	45	65	65	45	45	65
Synchronizing time (max):									
30° ± 2° displacement	Seconds	---	---	---	1	1	---	---	---
177° ± 2° displacement	Seconds	---	---	---	2	2	---	---	---

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3.5 Part number identification.- Synchros listed in Table I shall be identified by the following part numbers (see 6.2).

Part No.	Type	Item Name
7650633	15CX4	Synchro, Transmitter
7674777	15TR4a	Synchro, Receiver
7674778	15TDX4a	Synchro, Differential Transmitter
7674780	15CX4a	Synchro, Transmitter
7674781	15CT4a	Synchro, Control Transformer
7674782	15CDX4a	Synchro, Differential Transmitter
8603977	15TX4a	Synchro, Transmitter
8603979	15TR4b	Synchro, Receiver

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Certification of conformance to qualification tests.- Certification of conformance to qualification tests as specified in 3.1 is authorized for this item in accordance with 4.2.1.

4.2.1 Suppliers responsibilities.- The supplier shall be responsible for performance of all qualification tests and inspections necessary to insure that the synchros have met the qualification tests of MIL-S-20708, with exceptions as specified herein, under the following conditions:

a. Qualification tests and inspections shall be performed and quantitative test results submitted for a minimum of four (4) synchros produced to the contract or within one (1) year prior to the contract. The synchros tested and inspected shall conform with the design, construction, and physical dimensions specified in Figures 1, 2, and 3 as applicable.

b. The certification shall reference the contract or purchase order number and shall include part number identification, nomenclature, quantity tested, and company name. The certification shall also provide a statement as follows:

"As the authorized representative of the supplier, the undersigned warrants and represents that: All of the information supplied herewith is true and accurate and meets requirements for certification of conformance to qualification tests of MIL-S-20708/93(MU)."

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4.3 Quality conformance inspection.- Quality conformance examinations and tests shall be performed on the synchros in accordance with quality conformance tests of MIL-S-20708 as applicable and as specified herein.

4.4 Test methods and procedures.

4.4.1 Impedance.- Not applicable.

4.4.2 Altitude.- Not applicable.

4.4.3 Vibration.- Synchros shall be mounted rigidly to a vibration stand, shaft axis vertical and shaft extension downward, and shall be vibrated simultaneously in the vertical and horizontal directions for 8 hours ± 15 minutes, at 0.10 ± 0.02 inch amplitude and at a frequency varying linearly from 10 cps to 60 cps over a period of one minute. An acceptable test method is as follows: Mount the synchro with the shaft extension downward and shaft axis at 45° to the horizontal and vibrate in the vertical direction with a resultant amplitude yielding 0.10 ± 0.02 inch components parallel and perpendicular to the shaft axis.

4.4.4 Shock.- The shock test of MIL-S-20708 shall apply except that the shock blow and its application shall be in accordance with Method 202 of MIL-STD-202.

4.4.5 Ambient temperature.- The ambient temperature tests of MIL-S-20708 shall apply except that all synchros covered by this specification shall be subjected to a high temperature test of 85°C in lieu of 125°C.

4.4.6 Endurance.- The endurance test of MIL-S-20708 shall apply except that the total time of test shall be 1000 ± 10 hours and the time of test at 23 ± 5°C shall be 840 ± 4 hours.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with MIL-S-20708.

6. NOTES

6.1 Intended use.- Synchros covered by this specification (see 1.1) are intended only for maintenance support of in-service materiel. For synchros applicable to new materiel design see MIL-STD-710.

6.2 Ordering data.- MIL-S-20708 ordering data coverage shall apply except the synchro part number identification shall be in accordance with 3.5.

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6.3 Internal synchro design.- Figures 1 through 3 inclusive and Section 3 constitute the necessary Government requirements for supplying synchros covered by this specification. Details of internal design are optional. US Army Munitions Command detail drawings formerly used in procurement of these synchros are no longer required.

6.4 Information.- Information regarding synchros covered by this specification may be obtained from Commanding Officer, Frankford Arsenal, ATTN: SMUFA-T1000, Philadelphia, Pa. 19137.

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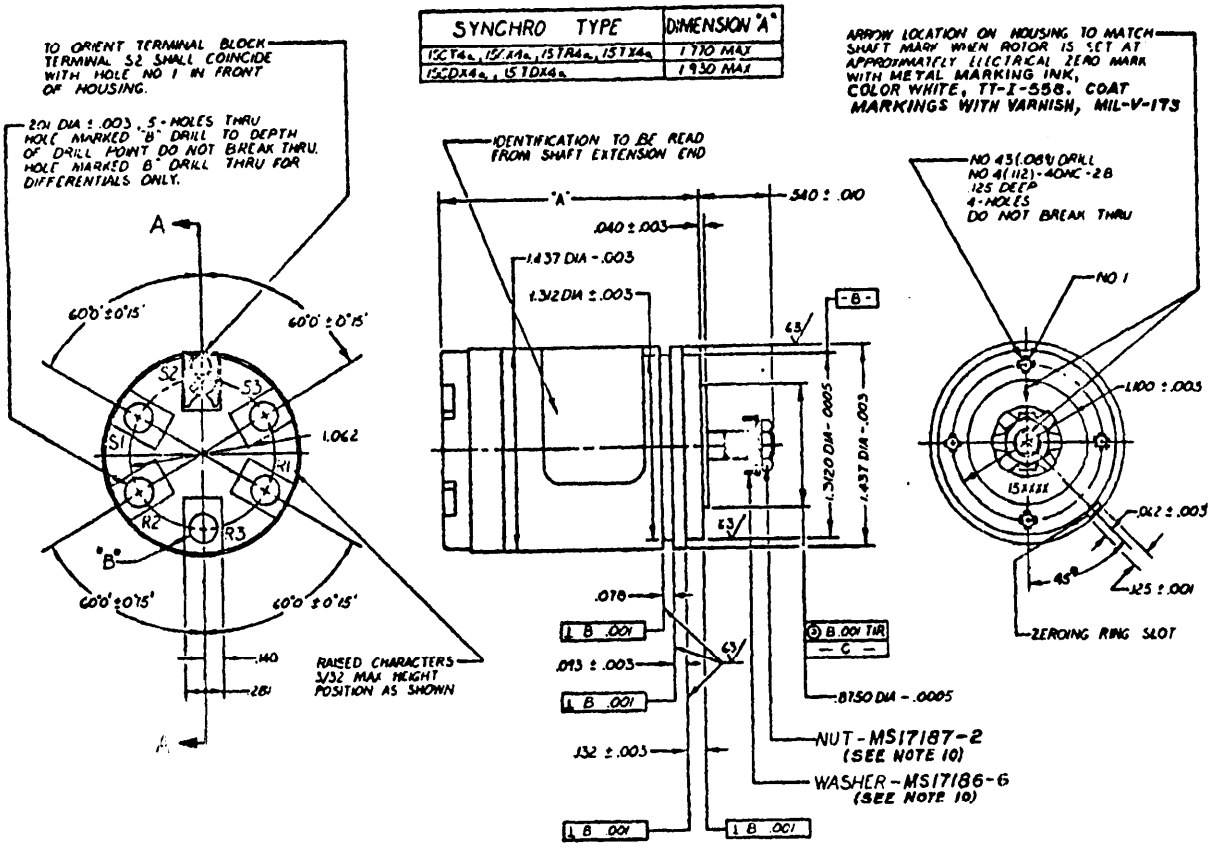
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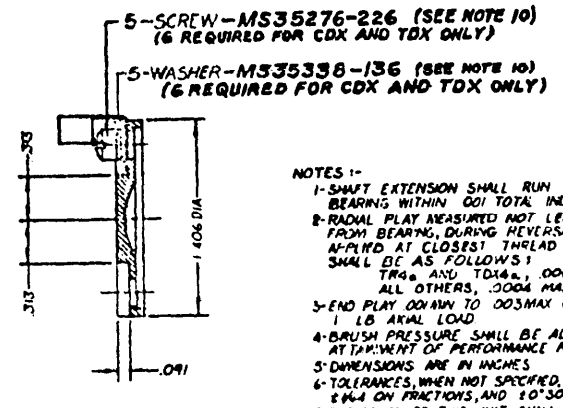
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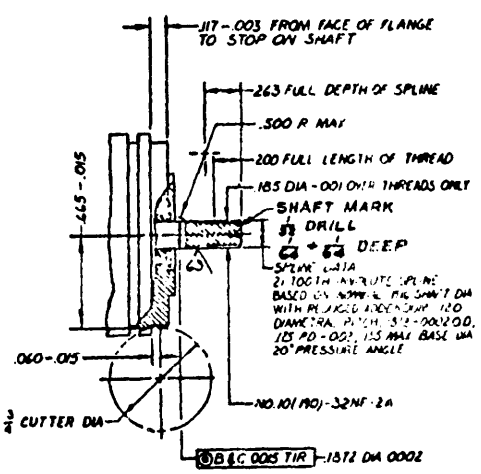
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PARTIAL SECTION A-A SHOWING TERMINAL RING COVER

NOTES :-
1- SHAFT EXTENSION SHALL RUN TRUE IN SHAFT BEARING WITHIN .001 TOTAL INDICATOR READING.
2- RADIAL PLAY MEASURED NOT LESS THAN 1/4 INCH FROM BEARING, DURING REVERSAL OF 1/2 LB LOAD APPLIED AT CLOSEST THREAD TO THE BEARING, SHALL BE AS FOLLOWS:
TR4a AND TDX4a, .0006 MAX TIR.
ALL OTHERS, .0004 MAX TIR
3- END PLAY .001 MIN TO .003 MAX UPON REVERSAL OF 1 LB AXIAL LOAD
4- BRUSH PRESSURE SHALL BE ADJUSTED TO PERMIT ATTAINMENT OF PERFORMANCE REQUIREMENTS.
5- DIMENSIONS ARE IN INCHES
6- TOLERANCES, WHEN NOT SPECIFIED, ARE ± .005 ON DECIMALS, ± 1/64 ON FRACTIONS, AND ± 0°30' ON ANGLES.
7- THE DESIGN OF THIS UNIT SHALL PROVIDE FOR THE TERMINAL BLOCK TO REMAIN FIXED WHEN TERMINAL SCREWS AND TERMINAL LUGS ARE REMOVED.
8- BREAK ALL SHARP EDGES .005 X 45°.
9- 125/ ALL OVER EXCEPT AS NOTED.
10- THE REQUIRED HARDWARE SHALL BE PACKAGED IN A VENTILATED ENVELOPE MADE OF MATERIAL HAVING A PH VALUE OF 6-8 AND COMPATIBLE WITH THE WEIGHT OF THE CONTENTS, THEN THE PACKAGED HARDWARE SHALL BE PLACED IN THE UNIT SYNCHRO PACKAGE.



PARTIAL SECTION SHOWING SPLINE SHAFT EXTENSION

FIGURE 1 · SYNCHROS, 400 CYCLE, SIZE 15

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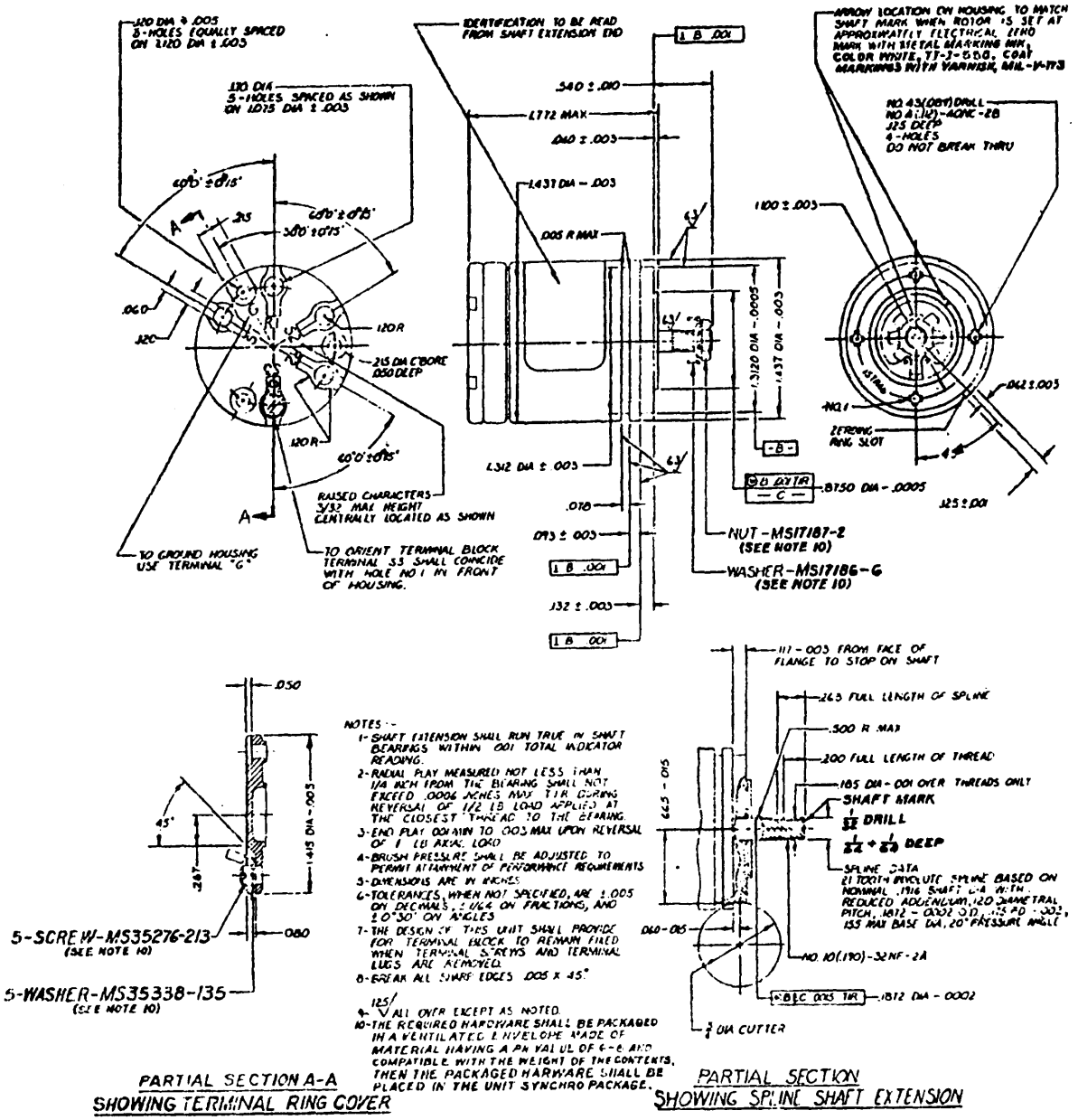
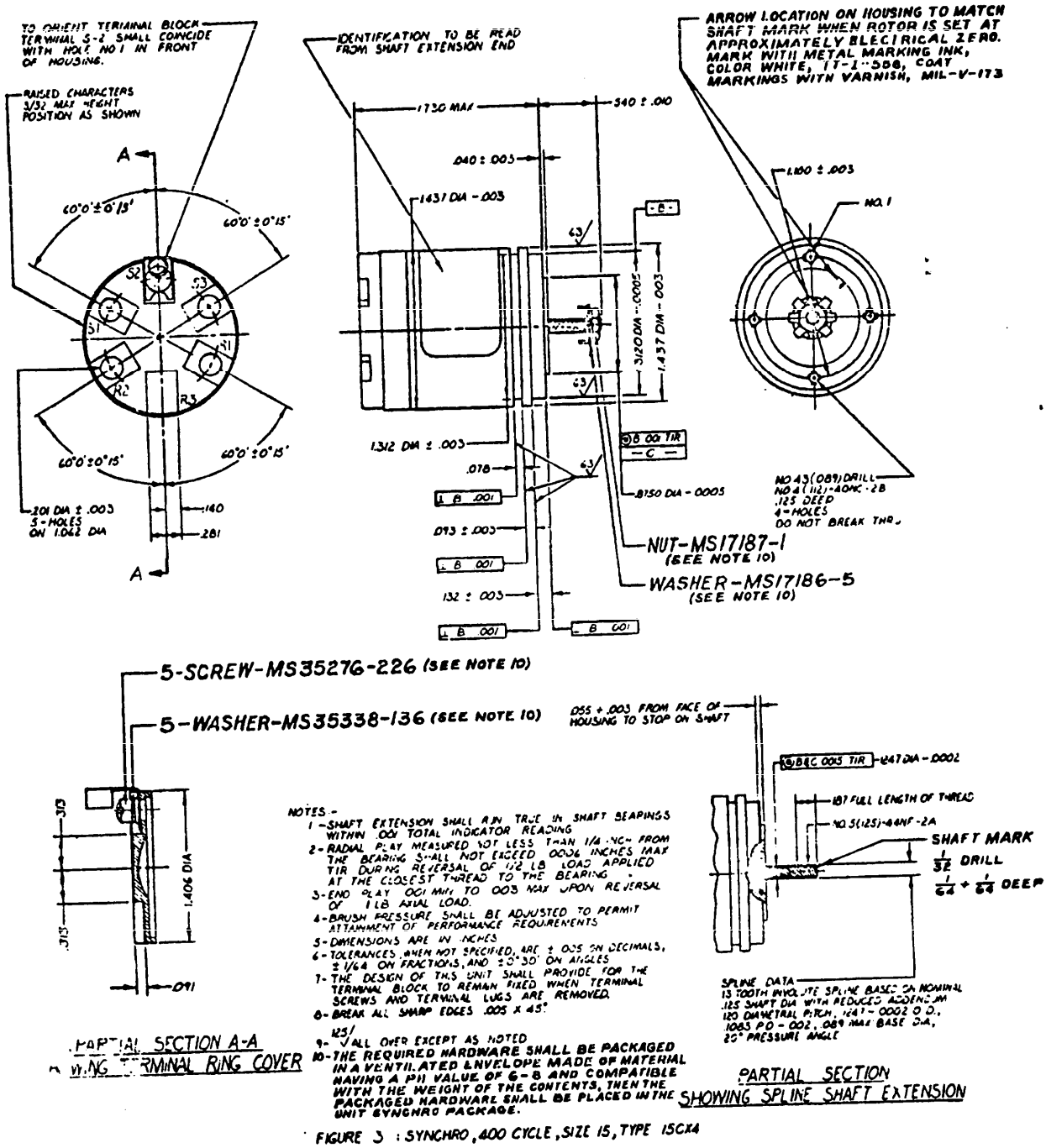


FIGURE 2 : SYNCHRO, 400 CYCLE, SIZE 15, TYPE 15TR4b

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PARTIAL SECTION A-A
WIRING TERMINAL RING COVER

NOTES:
1 - SHAFT EXTENSION SHALL RUN TRUE IN SHAFT BEARINGS WITHIN .001 TOTAL INDICATOR READING
2 - RADIAL PLAY MEASURED NOT LESS THAN 1/4 INCH FROM THE BEARING SHALL NOT EXCEED .0006 INCHES MAX TIR DURING REVERSAL OF 1/2 LB LOAD APPLIED AT THE CLOSEST THREAD TO THE BEARING.
3 - END PLAY .001 MIN TO .003 MAX UPON REVERSAL OF 1 LB AXIAL LOAD.
4 - BRUSH PRESSURE SHALL BE ADJUSTED TO PERMIT ATTAINMENT OF PERFORMANCE REQUIREMENTS
5 - DIMENSIONS ARE IN INCHES
6 - TOLERANCES WHEN NOT SPECIFIED, ARE ± .005 ON DECIMALS, ± 1/64 ON FRACTIONS, AND ± 0°30' ON ANGLES
7 - THE DESIGN OF THIS UNIT SHALL PROVIDE FOR THE TERMINAL BLOCK TO REMAIN FIXED WHEN TERMINAL SCREWS AND TERMINAL LUGS ARE REMOVED.
8 - BREAK ALL SHARP EDGES .005 X 45°
9 - 125/ ALL OVER EXCEPT AS NOTED
10 - THE REQUIRED HARDWARE SHALL BE PACKAGED IN A VENTILATED ENVELOPE MADE OF MATERIAL HAVING A PH VALUE OF 6-8 AND COMPATIBLE WITH THE WEIGHT OF THE CONTENTS, THEN THE PACKAGED HARDWARE SHALL BE PLACED IN THE UNIT SYNCHRO PACKAGE.

PARTIAL SECTION SHOWING SPLINE SHAFT EXTENSION

FIGURE 3 : SYNCHRO, 400 CYCLE, SIZE 15, TYPE 15CX4

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